# MAHATMA GANDHI UNIVERSITY, KOTTAYAM
## MGU-UGP (HONOURS)
## THIRD SEMESTER EXAMINATION
## Model Question Paper
## MG3VACCHE200 – FORENSIC CHEMISTRY
(2024 ADMISSION ONWARDS)

**Duration: 1.5 HRS** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

## Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. Which organ is primarly affected in lead poisoning [U] [CO:1]
   A. Kidney B. Heart C. Lungs D. Liver

2. The first step in the treatment of poisoning is to [A] [CO:1]
   A. Administrate antidote B. Identify the poison
   C. Gastric Lavage D. Airway, Breathing and circulation

3. Which antidote is used in Cyanide poisoning? [K] [CO:1]
   A. Atropine B. EDTA C. Sodium thiosulphate D. Naloxone

4. What is the primary explosive component in gelatine sticks used in human bombs? [U] [CO:2]
   A. Ammonium Nitrate B. RDX C. Nitroglycerin D. TNT

5. RDX is a powerful explosive. What does RDX stand for? [U] [CO:2]
   A. Reactive Detonating Xenon B. Rapid Detonation Explosive
   C. Research Development Explosive D. Royal Demolition Explosive

6. What is the name of the black residue left by gunpowder after firing? [U] [CO: 2]
   A. Gun metal B. Bullet grease C. Gunshot residue (GSR) D. Metal shavings

7. In traced forgery, how is the signature produced? [U] [CO: 3]
   A. From memory B. With a different writing instrument
   C. By placing transparent paper over the genuine signature and copying it
   D. Using ink erasure

8. Which of the following is a confirmatory test for urine in forensic analysis? [U] [CO:4]
   A. Iodine test B. Phadebas test
   C. DMAC (p-Dimethylaminocinnamaldehyde) test D. Teichmann test

9. What is DNA fingerprinting primarily used for in forensic science? [A] [CO:4]
   A. Measuring blood pressure
   B. Identifying handwriting styles
   C. Identifying individuals based on their genetic material
   D. Determining age of bones

**[1x9 = 9]**

## Part B

Short Answer Questions

Answer any **5** questions

Each question carries **4** marks

10. Describe the classification of poisons based on their mode of action [U] [CO:1]
11. Write a note on neutron activation analysis [U] [CO:1]
12. Describe the composition of a bullet [U] [CO:2]
13. Explain the working of a metal detector [U] [CO:2]
14. Discuss the inherent signs of forged signatures [U] [CO: 3]
15. Explain the types of fractures that occur in glass during impact [A] [CO: 4]
16. What is DNA finger printing [U] [CO: 4]

**[4x5 = 20]**

## Part C

Short Essay Questions

Answer any **3** questions

Each question carries **7** marks

17. Explain the general treatments and antidotes provided for a poisoning case. [U] [CO:1]
18. Explain the detection of accidental explosion due to fire works and matches. [U] [CO:2]
19. Discuss the different types of signature forgeries [K] [CO: 3]
20. Explain the various methods used to detect the purity of gold [U] [CO: 3]
21. What do you mean by traces and tracks? Explain. [U] [CO: 4]

**[7x3 = 21]**